kader Academy

Machine Safety Specialist (CE)

Tab 1 – Module 2: Machinery Directive

TAB1: principles machine safety
Tab1-2/5 20231010

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Tab1-2/5-Sh.2

Machinery Directive (2006/42/EG)

Machinery directive 89/392/EEC
- Mandatory from Januari 1995

Machinery directive 98/37/EG
- Mandatory from Augustus 1998

Current Machinery directive 2006/42/EG
- **Mandatory from December 2009**

Regulation on Machinery Products (successor to 2006/42/EC)

- *Concept published*
- *Planned publication and entry into force Q4 2022 / Q1 2023*
- *Expected transition period of 30 months (after entry into force)*

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Tab1-2/5-Sh.3

Machinery Directive – Scope

(Machinery Directive, Article 1 (1))

Applies to:

a. Machinery

b. Interchangeable equipment

c. Safety components

Tab1-2/5-Sh.4

Machinery Directive – Scope

(Machinery Directive, Article 1 (1))

Applies to:

d. Lifting accessories

e. Chains, ropes and webbing

f. Removable mechanical transmission devices

g. Partly completed machinery

Definitions

(Machinery Directive, Article 2)

Basic definition machinery:

- an **assembly**, fitted with or intended to be fitted with a drive system other than directly applied human or animal effort, **consisting of linked parts or components, at least one of which moves, and which are joined together for a specific application;**

Definitions

(Machinery Directive, Article 2)

Machinery (continuation)

- an assembly referred to in the first indent, missing only the components to connect it on site or to sources of energy and motion;
- an assembly referred to in the first and second indents, ready to be installed and able to function as it stands only if mounted on a means of transport, or installed in a building or a structure;

Definitions
(Machinery Directive, Article 2)

Machinery (continuation)

- Assemblies of machinery referred to in the first, second and third indents or partly completed machinery referred to in point (g) which, in order to achieve the same end, are arranged and controlled so that they function as an integral whole;

On the basis of the Guide to the Application of the Machinery Directive, all three of the following criteria must be fulfilled in order to constitute an assembly of machinery:

1. *The constituent units are joined together in order to fulfil a common function, for example the production of a particular product;*
2. *The constituent units are functionally linked in such a way that the operation of each unit directly affects the operation of other units or of the assembly as a whole, so that a risk assessment is required for the whole assembly;*
3. *The constituent units share a common control system*

A group of machines that are connected to each other but work independently is not considered to be an assembly of machinery in the above sense.

Definitions
(Machinery Directive, Article 2)

Machinery (continuation)

- an assembly of linked parts or components, at least one of which moves, and which are joined together, intended for lifting loads and whose only power source is directly applied human effort;

Question 1-4

Which of the following examples is considered machinery?

- Production line with overarching control
- Lathe
- Press brake
- Jack (car)

Definitions

(Machinery Directive, Article 2)

Interchangeable equipment

- Machinery function is changed
- Placed on the market to be assembled with a machine or tractor by the operator himself
- Equipment is not a tool

Definitions

(Machinery Directive, Article 2)

Safety Component

- Fulfils a safety function
- Placed on the market separately
- The failure and/or malfunction of which endangers the safety of persons
- Not necessary for the operation of the machinery

Annex V: list of 'example' safety components;

Definitions

(Machinery Directive, Article 2)

Lifting accessory

- Suitable for lifting or hoisting a load
- Not attached to the lifting machinery
- Placed between the machinery and the load or on the load itself or which is intended to constitute an integral part of the load
- Independently placed on the market

Question 1-5

- Do the following products fall within the definition of Lifting accessories?
- Which of the products must be provided with CE marking?

Spreader beam

Jack

Four-leg sling

Manual chain hoist

Definitions

(Machinery Directive, Article 2)

Chains, ropes and webbing

- designed and constructed for lifting purposes
- as part of lifting machinery or lifting accessories

Definitions
(Machinery Directive, Article 2)

Removable mechanical transmission device
- Removable
- For transmitting power
- Equipment with the guard = one product

Definitions
(Machinery Directive, Article 2)

Partly completed machinery
- Is almost machinery
- Cannot in itself perform a specific application
- Only intended to be incorporated or assembled
- No CE marking

Example from machinery directive:
- A drive system.

Question 1-6

- Is it necessary to apply a CE marking to a conveyor belt (newly placed on the market) based on the Machinery Directive?

Machinery Directive - Scope

(Machinery Directive, Article 1 (1))

Exclusions 1/3 :

- Specific equipment for use in fairgrounds and/or amusement parks
- Machinery specially designed or put into service for nuclear purposes
- Weapons, including firearms
- Seagoing vessels and mobile offshore units and machinery installed on board such vessels and/or units
- Machinery specially designed and constructed for military or police purposes
- Mine winding gear
- Machinery intended to move performers during artistic performances

Machinery Directive – Scope

(Machinery Directive, Article 1 (1))

Exclusions (2/3):

- Electrical and electronic products falling within the following areas, insofar as they are covered by Council Directive 73/23/EEC of 19 February 1973 on the harmonization of the laws of Member States relating to electrical equipment designed for use within certain voltage limits:
  - household appliances intended for domestic use
  - audio and video equipment
  - information technology equipment
  - ordinary office machinery
  - low-voltage switchgear and control gear
  - electric motors
- the following types of high-voltage electrical equipment:
  - switch gear and control gear
  - transformers

Machinery Directive – Scope

(Machinery Directive, Article 1 (1))

Exclusions (3/3):

- The following means of transport (except machinery fitted to such vehicles)):
  - agricultural and forestry tractors for the risks covered by Directive 2003/37/EC;
  - motor vehicles and their trailers covered by Council Directive 70/156/EEC;
  - motor vehicles exclusively intended for competition;
  - means of transport by air, on water and on rail networks with the exclusion of machinery mounted on these means of transport;
  - vehicles covered by Directive 2002/24/EC of the European Parliament and of the Council of 18 March 2002 relating to the type-approval of two or three-wheel motor vehicles;

Machinery Directive – Scope

Exercise

- An EC Declaration of Conformity for an electric bicycle (see below). Is this correct, or does it fall under the vehicle exception?

DEZE ELEKTRISCHE FIETS IS GEPRODUCEERD CONFORM EN IN GEMEENSCHAP VAN DE RICHTLIJNEN VAN DE EUROPESE UNIE AANGAANDE ELEKTRISCHE FIETSEN MET EEN MAXIMUM SNELHEID <= 25 KM/UUR EN EEN MAXIMUM (MOTOR) VERMOGEN VAN 250W

DE VOLGENDE "EN" (ISO) NORMEN ZIJN VAN TOEPASSING:

EN ISO 12100:2010; EN 15194:2009; EN 14764:2005; EN 349:1993+A1:2008; EN 953:1997+A1:2009; EN 13857:2008

DE EG RICHTLIJNEN AANGAANDE GOEDKEURING:

2006/42/EC MACHINERY DIRECTIVE; 2006/95/EC LOW VOLTAGE DIRECTIVE; 2004/108/EC ELECTROMAGNETIC COMPATIBILITY DIRECTIVE

RICHTLIJN ELEKTROMAGNETISCHE COMPATIBILITEIT (2014/30/EU), MACHINERICHTLIJN (2006/42/EG)

UITVOERENDE INSTANTIE EG CERTIFICERING: ENTE CERTIFICAZIONE MACHINE S.R.L., VIA MINCIO 386, 41056 SAVIGNANO, ITALIE; AANGEMELDE INSTANTIE EG-TYPE ONDERZOEK NUMMER: 1282

CERTIFICAAT NUMMER: 111123/ZJV845

CERTIFICAAT DATUM: NOVEMBER 2011

Machinery Directive – Scope

Exercise

- An EC Declaration of Conformity for an electric bicycle (see below). Is this correct, or does it fall under the vehicle exception?

Article 2

Scope

1. This Regulation shall apply to all two- or three-wheel vehicles and quadricycles as categorised in Article 4 and Annex I ('L-category vehicles'), that are intended to travel on public roads, including those designed and constructed in one or more stages, and to systems, components and separate technical units, as well as parts and equipment, designed and constructed for such vehicles.

This Regulation also applies to enduro motorcycles (L3e-AxE (x = 1, 2 or 3)), trial motorcycles (L3e-AxT (x = 1, 2 or 3)) and heavy all terrain quads (L7e-B) as categorised in Article 4 and Annex I.

2. This Regulation does not apply to the following vehicles:

(a) vehicles with a maximum design speed not exceeding 6 km/h;

(b) vehicles exclusively intended for use by the physically handicapped;

(c) vehicles exclusively intended for pedestrian control;

(h) pedal cycles with pedal assistance which are equipped with an auxiliary electric motor having a maximum continuous rated power of less than or equal to 250 W, where the output of the motor is cut off when the cyclist stops pedalling and is otherwise progressively reduced and finally cut off before the vehicle speed reaches 25 km/h;

Quote from **REGULATION (EU) No 168/2013** Article 1 concerning the approval of two- or three-wheel motor vehicles.

Machinery Directive – Scope

Exercise

- Does the same apply for an E-bike with the following specifications?

Modeljaar	2020
Framemaat	M, L
Juizz maat	1.60 – 1.70 (M), 1.70 – 1.80 (L)
Kleur	Dutch Orange, Matte Black
Type versnelling	Derailleur
Versnelling omschrijving	Shimano Derailleur SLX 11 Speed
Type rem	Hydraulische schijfrem

Ondersteunt tot	45km/h
Frame type Juizz	Heren, Mixed
Gewicht in kg	26,8
Motor positie	Achterwiel
Motor omschrijving	QWIC 400 Watt achterwielmotor 45Nm max 45 km/h

'Dangerous' machinery

(Machinery Directive, Article 12, (3 and 4))

Listed in Annex IV of the Machinery Directive, e.g.:

- Portable chainsaws
- Woodworking machinery
- Presses; stroke > 6 mm and speed > 30 mm/s.
- Injection or compression moulding machinery
- Machinery for underground working
- Trucks for the collection of household garbage
- Removable mechanical transmission devices
- Vehicle servicing lifts
- Devices for the lifting of persons or of persons and goods involving a hazard of falling from a vertical height of more than three metres.
- Protective devices designed to detect the presence of persons
- Logic units to ensure safety functions
- Roll-over protective structures (ROPS)
- Falling-object protective structures (FOPS)

'Dangerous' machinery

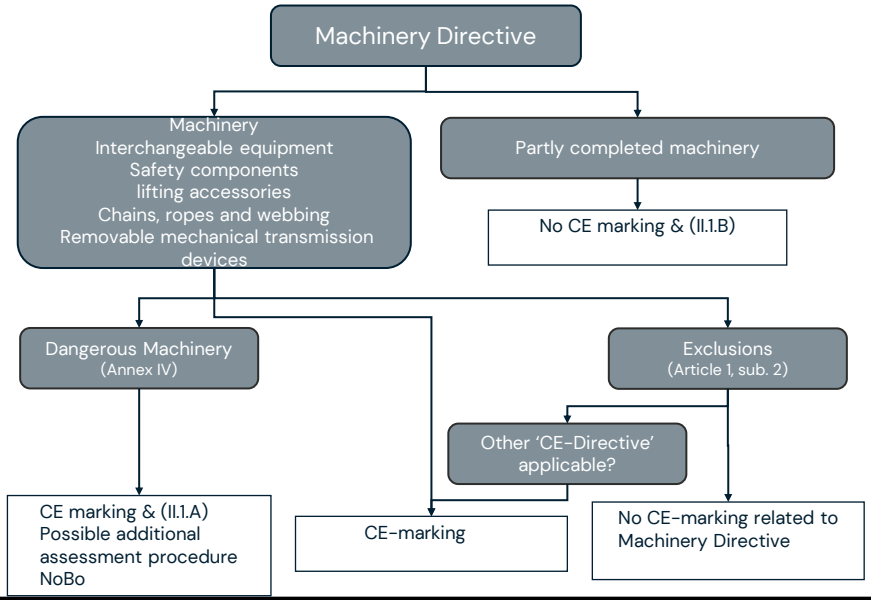
(Machinery Directive, Article 12, (3 and 4))

Conformity assessment procedures

1. Manufacture according to harmonized (type-C) standard and essential requirements is fully covered by that standard:
   - Conformity assessment by manufacturer (no Notified Body) or
   - EC type-examination (Notified Body) or
   - Full quality assurance (Notified Body)

2. Manufacture not in accordance with harmonized (type-C) standard and/or essential requirements are not fully covered (usually requires involvement of a Notified Body):
   - EC type-examination (Machinery Directive, Annex IX)
   - Full quality assurance (Machinery Directive, Annex X).

Machinery Directive – Scope (summarized)



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